

LABORATORY STRUCTURE

WAFT | Workspace Organization

Overview

A WAFT **Laboratory** is a directory containing everything needed for agent evolution. Understanding the structure helps you work effectively.

Directory Layout

```
my_lab/
├── waft.toml           # Configuration
├── agents/            # Agent source code
│   ├── __init__.py
│   ├── base_agent.py
│   └── custom_agents/
├── prompts/          # System prompts
│   ├── default.txt
│   └── specialized/
├── mutations/         # Mutation configs
│   └── standard.json
├── checkpoints/       # Evolution snapshots
│   └── gen_050/
├── seeds/             # Agent state saves
│   └── MyAgent/
├── flight_data/       # Telemetry database
│   └── flight.db
├── results/           # Output files
│   ├── reports/
│   └── exports/
└── logs/              # Application logs
```

Key Files

waft.toml

Main configuration:

```
[project]
name = "my_lab"
version = "0.1.0"

[evolution]
population_size = 20
mutation_rate = 0.1

[gym]
timeout_seconds = 300

[storage]
database = "flight_data/flight.db"
```

agents/

Your agent source code:

```
# agents/my_agent.py
from waft import Agent

class MyAgent(Agent):
    system_prompt = "...

    def process(self, input: str) -> str:
        return self.llm.generate(...)
```

prompts/

Reusable system prompts:

```
prompts/
├─ default.txt      # Standard prompt
├─ code_review.txt  # For code agents
├─ research.txt     # For research agents
└─ safety_focused.txt # High safety emphasis
```

Working with Laboratories

Create

```
waft new my_lab
cd my_lab
```

Verify

```
waft verify
waft verify --fix # Auto-fix issues
```

Status

```
waft status
```

Output:

```
Laboratory: my_lab
Agents: 3
Generations: 47
Best Fitness: 0.823
Last Run: 2 hours ago
```

Multiple Laboratories

You can have multiple labs for different experiments:

```
waft new experiment_a
waft new experiment_b

# Work in specific lab
cd experiment_a
waft evolve ...

# Or specify lab
waft --lab experiment_b evolve ...
```

Backup & Export

Backup entire lab

```
waft lab backup --output my_lab_backup.zip
```

Export just results

```
waft lab export --results-only
```

Clone lab

```
waft lab clone my_lab my_lab_copy
```

LABORATORY STRUCTURE | Organization Matters